Ruben Chenevat

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Québec, Canada · August 2026

Research interests: Optimal control and dynamical systems — Decision-making under constraints and uncertainty — Biological and environmental modelling

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Current position

Postdoctoral Researcher — **École d'Actuariat, Université Laval**
Stochastic optimal control of constrained and jump systems, HJB equations, averaged cost control
Québec, Canada
Feb 2026–Sep 2026

Professional experience

Research Engineer — **G-EAU, INRAE**
Modelling and optimal control of agrivoltaic systems
Montpellier, France
Nov 2025–Jan 2026

Education

PhD in Mathematics and Modelling — **Université de Montpellier, MISTEA, INRAE**
"Optimal control of irrigation: double mathematical and agronomic modeling towards an application to the Optirrig model"
Supervisors: Alain Rapaport, Bruno Cheviron, Sébastien Roux
Montpellier, France
Oct 2022–Oct 2025

Master's degree in Mathematics — **Université Paris-Saclay**
Analysis, Modelling, Simulation
Gif-sur-Yvette, France
Sep 2021–Aug 2022

Normalien student in Mathematics — **École Normale Supérieure de Rennes**
Civil servant trainee status
Agrégation de Mathématiques (session 2021)
Rennes, France
Sep 2018–Aug 2022

Research internships and visits

Visit – Optimal control problems with state constraints for crop irrigation — SYSTEC, FEUP
Collaborator: Maria do Rosário de Pinho
Porto, Portugal
Jan 2025–April 2025

Internship – Unicity and minimality of vortex solutions in variational models — IMT, Université Paul Sabatier
Supervisor: Radu Ignat
Toulouse, France
April 2022–Aug 2022

Internship – Mathematical problems in the analysis of the stability and the structure of matter — QMATH, University of Copenhagen
Supervisor: Jan-Philip Solovej
Copenhagen, Denmark
May 2019–July 2019

Publications

Journal Articles

[A2] About the bang-bang principle for controlled affine dynamics with Brownian noise.
Ruben Chenevat, Dan Goreac, Qinlong Li, Alain Rapaport.
Journal of Convex Analysis, 33, 801–816 (2026).

[A1] Optimal structures of crop irrigation strategies with state constraints.
Ruben Chenevat, Bruno Cheviron, Sébastien Roux, Alain Rapaport.
Journal of Optimization Theory and Applications, 208, 18 (2026). DOI: 10.1007/s10957-025-02854-7

Conference Papers

[C3] Extremal stochastic controls for affine jump systems with applications to two-line insurance models.
Hezhen Bao, Ruben Chenevat, Dan Goreac, Juan Li.
15th Asian Control Conference (ASCC), Bali, Indonesia, 2026.

[C2] Common structures of optimal solutions for a crop irrigation problem under various constraints and criteria.
Ruben Chenevat, Bruno Cheviron, Sébastien Roux and Alain Rapaport.
63rd IEEE Conference on Decision and Control (CDC), Milan, Italy, 2024. DOI: 10.1109/CDC56724.2024.10886226

[C1] About the bang-bang principle for piecewise affine systems.
Ruben Chenevat, Bruno Cheviron, Sébastien Roux and Alain Rapaport.
63rd IEEE Conference on Decision and Control (CDC), Milan, Italy, 2024. DOI: 10.1109/CDC56724.2024.10886617

Preprints and Submitted Papers

- Extension of the bang-bang principle for piecewise affine dynamics under state constraints.
  Ruben Chenevat, Bruno Cheviron, Sébastien Roux, Alain Rapaport.
  (Submitted, April 2026).

Talks in international conferences

"Optimizing crop irrigation under biological and operational constraints with meteorological uncertainty"
EUROGEN 2025, Lahti, Finland, Sep 2025.

"Common structures of optimal solutions for a crop irrigation problem under various constraints and criteria"
CDC 2024, Milan, Italy, Dec 2024.

"About the bang-bang principle for piecewise affine systems"
CDC 2024, Milan, Italy, Dec 2024.

"About the bang-bang principle in a non-smooth setting"
FGS 2024, Gijon, Spain, June 2024.

Poster: "Optimal structures for a model of crop irrigation with constraints"
SMAI-MODE 2024, Lyon, France, March 2024.

Teaching

Tutorials – Logic and set theory — Université de Montpellier, Fall 2023
First-year undergraduate level

Lab sessions – Numerical analysis of ODEs and PDEs — Université de Montpellier, Spring 2023
Third-year undergraduate level

Tutorials – Analysis I: Functions of one variable and sequences — Université de Montpellier, Fall 2022 & 2023
First-year undergraduate level

Tutorials – Calculus — Université de Montpellier, Fall 2022 & 2023
First-year undergraduate level

Grants

Hubert Curien partnership France-Portugal (Program PHC-PESSOA 2025-2027)
Project: "OITE: Optimizing Irrigation efficiency under Technical and Environmental constraints"

Scientific activities

Project "BOUM" SMAI 2025
Funding for the organization of a workshop day on Optimisation, Modelling and Control (held on June 12, 2025)
Co-organized with Anas Bouali and Gildas Dadjo

Working group (April 2023–July 2025)
"Optimal Control & Pontryagin Maximum Principle"

Academic service

Reviewer for:

IFAC World Congress 2026

Awards and scholarships

PhD label: International research school of Agreenium (EIR-A 2024)

PhD label: Digital agriculture convergence lab (#DigitAg 2023)

Skills

Languages: French (native), English (fluent), German (basic), Portuguese (basic)
Programming: Python, Julia, R, MATLAB
Optimisation software: BOCOP, Control Toolbox
Typesetting: LaTeX